

Computing Systems for Autonomous Driving: State-of-the-Art and Challenges

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Abstract— The recent proliferation of computing technologies (e.g., sensors, computer vision, machine learning, and hardware acceleration), and the broad deployment of communication mechanisms (e.g., DSRC, C-V2X, 5G) have pushed the horizon of autonomous driving, which automates the decision and control of vehicles by leveraging the perception results based on multiple sensors. The key to the success of these autonomous systems is making a reliable decision in real-time fashion. However, accidents and fatalities caused by early deployed autonomous vehicles arise from time to time. The real traffic environment is too complicated for current autonomous driving computing systems to understand and handle. In this paper, we present state-of-the-art computing systems for autonomous driving, including seven performance metrics and nine key technologies, followed by twelve challenges to realize autonomous driving. We hope this paper will gain attention from both the computing and automotive communities and inspire more research in this direction.

I. INTRODUCTION

Recently, with the vast improvements in computing technologies, e.g., sensors, computer vision, machine learning, hardware acceleration, and the wide deployment of communication mechanisms, e.g., Dedicated short-range communications (DSRC), Cellular Vehicle-to-Everything (C-V2X), and 5G, autonomous driving techniques have attracted massive attention from both the academic and automotive communities [1], [2]. According to [3], the global autonomous driving market expects to grow up to \$173.15B by 2030. Many automotive companies have made enormous investments in this domain, including ArgoAI, Audi, Baidu, Cruise, Mercedes-Benz, Tesla, Uber, and Waymo, to name a few [4], [5], [6], [7]. Several fleets of SAE (Society of Automotive Engineers) L4 vehicles have been deployed around the world [8], [9].

To achieve autonomous driving, determining how to make the vehicle understand the environment correctly and make safe controls in real-time is the essential task. Rich sensors including camera, LiDAR (Light Detection and Ranging), Radar, Inertial Measurement Unit (IMU), Global Navigation Satellite System (GNSS), and Sonar, as well as powerful computation devices, are installed on the vehicle [10], [11], [12], [13], [14]. This design makes autonomous driving a real powerful "computer on wheels." In addition to hardware, the rapid development of deep learning algorithms in object/lane detection, simultaneous localization and mapping (SLAM), and vehicle control also promotes the real deployment and prototyping of autonomous vehicles [15], [16], [17], [18]. The

autonomous vehicle's computing systems are defined to cover everything (excluding the vehicle's mechanical parts), including sensors, computation, communication, storage, power management, and full-stack software. Plenty of algorithms and systems are designed to process sensor data and make a reliable decision in real-time.

However, news of fatalities caused by early developed autonomous vehicles (AVs) arises from time to time. Until August 2020, five self-driving car fatalities happened for level-2 autonomous driving: four of them from Tesla and one from Uber [19]. Table I summarizes the date, place, company, and reasons for these five fatalities. The first two fatalities attributed to Tesla happened in 2016 with the first accident occurring because neither the Autopilot system nor the driver failed to recognize the truck under thick haze. The vehicle in the second incident mistook the truck for open space. In the 2018 incident involving Tesla, the autopilot failed to recognize the highway divider and crashed into it. The most recent fatality from Tesla happened in 2019 because the vehicle failed to recognize a semi-trailer. The fatality from Uber happened because the autonomous driving system failed to recognize that pedestrians jaywalk.

In summary, all four incidents associated with Tesla are due to perception failure, while Uber's incident happened because of the failure to predict human behavior. Another fact to pay attention to is that currently, the field-testing of level 2 autonomous driving vehicles mostly happens in places with good weather and light traffic conditions like Arizona and Florida. The real traffic environment is too complicated for the current autonomous driving systems to understand and handle easily. The objectives of level 4 and level 5 autonomous driving require colossal improvement of the computing systems for autonomous vehicles.

This paper presents state-of-the-art computing systems for autonomous driving, including seven performance metrics and nine key technologies, followed by eleven challenges and opportunities to realize autonomous driving. The remaining parts of this paper are organized as follows: Section II discusses the reference architecture of the computing systems for autonomous driving. In Section III, we show some metrics used in the evaluation of the computing system. Section IV discusses the key technologies for autonomous driving. Section V presents possible challenges. Finally, this paper concludes in Section VI.